

MTH 131 Practice Exam 1: P.1-P.5**Name:****No calculators.** For full credit always show your work.[5 pts] Check *all* that apply**1.**

Number	Natural	Integer	Rational	Irrational	Real
5	☐	☐	☐	☐	☐
$\sqrt{2}$	☐	☐	☐	☐	☐
$\frac{5}{3}$	☐	☐	☐	☐	☐

[5 pts] Evaluate when $x = -1$ and $y = 3$.

2. $x^2 + 2x^3 - 2(2y - x)$

[5 pts each] Perform the operation.

3. $\frac{5}{18} \div \frac{10}{3} - \left(\frac{3}{2}\right)^2$

4. $\frac{10 - \sqrt{32}}{6}$

[5 pts each] Describe and correct the answer.

5. ~~$\sqrt{4+16} = \sqrt{4} + \sqrt{16} = 2 + 4 = 6$~~

6. ~~$5 - (x + 4) = 5 - x + 4 = -x + 9$~~

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[5 pts each] Simplify each expression. Write all exponents as positive.

7. $6z^2(-3z^3)^2$

8. $\frac{3^{-4}x^{-1}y^{-4}}{9^{-2}(x^{-1}y)^{-3}}$

9. $\frac{2x+4}{(x-3)^2} \times \frac{x^2-3x}{x^2+3x+2}$

10. $8y^2 - [2y^2 - (3y - 8)]$

11. $\frac{x(x+1)^{-1/2} + (x+1)^{1/2}}{(x+1)^{3/2}}$

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[5 pts] Find the product (expand).

12. $\left(2x - \frac{1}{3}\right)^2$

[5 pts] Factor out completely.

13. $2x^3 - 4x^2 - 16x$

14. $2y^2 + 5x + 3$

15. $9z^2 - 49$

16. $x^3 - 4x^2 + 2x - 8$

17. $2(t^2 - 3)^3 - 3(t^2 - 3)^2$

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[5 pts] Rewrite the fraction as a sum of terms with no denominator.

18. $\frac{3x^3 - x^2 + 8}{x^{3/2}}$

X [5 pts] Complete both parts of the problem.

19. A widget manufacturer can produce and sell x widgets per quarter. The total cost C (in dollars) for producing x widgets is $C = 1,000 + 100x$ and the total revenue R (in dollars) is $R = 200x$.

(a) Find the profit in terms of x .

(b) Find the profit obtained by selling 50 widgets.

X [5 pts] Find the domain of the expression. Write answers in interval notation.

20. $\sqrt{x-5}$